What is a consumer?

BACKGROUND

mammals (say 'ma-mels') – warm-blooded animals with backbones that feed their young with milk

producers (say 'pro-dews-ers') – living things that make their own food from non-living ingredients

Consumers are types of organisms that eat or use food products made by producers or that eat other consumers. Large land mammals, birds and reptiles are consumers.

There are three main types of consumers:

- **herbivores** (consumers that eat plants)
- **carnivores** (consumers that eat animals, or meat)
- **omnivores** (consumers that eat both plants and animals).

Humans are omnivores. We eat all sorts of foods produced by plants, and we also eat other animals that eat plants and animals.

☞ **1** List six types of living things that are consumers.

_____ _____

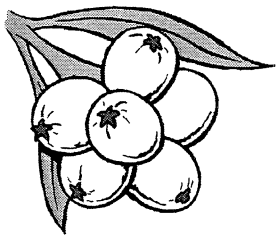
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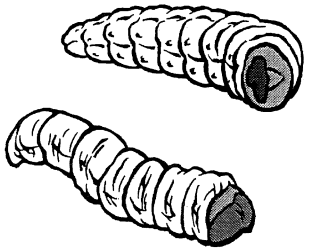
2 Decide whether these things are producers or consumers. Circle your response for each one.



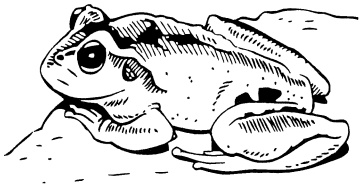
a producer OR consumer



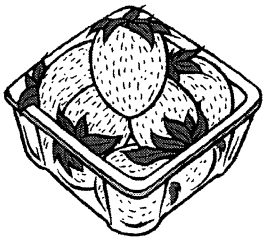
b producer OR consumer



c producer OR consumer



d producer OR consumer



e producer OR consumer

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What is a consumer? CONT'D

Most carnivores are **predators**. Predators hunt, kill and eat other living things. The living thing that a predator hunts is called its **prey**. For example, the African lion is a fierce predator that needs meat to survive. Its prey includes wildebeest, zebras and giraffes.

3 a) Which of the living things in the box are predators? Underline your choices.

b) Which of the living things in the box are prey for different types of predators? Circle your choices.

tiger	caterpillar	snail	zebra
brown bear	mouse	rabbit	dingo

c) Look at the living things you have underlined. List each one below and write whether that predator is a carnivore or omnivore.

Predator: _____ Carnivore or omnivore? _____

Predator: _____ Carnivore or omnivore? _____

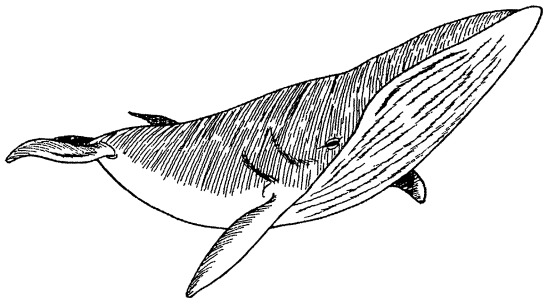
Predator: _____ Carnivore or omnivore? _____

4 a) Which of these animals have no natural predators to fear in their ecosystems? Circle your choices.

blue whale	lizard	prawn
rabbit	crocodile	lion

b) Look at the animals you have circled. What animal hunts and kills these animals, but is not considered a natural predator?

_____



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What is a decomposer?

BACKGROUND

nutrients (say 'nu-tree-ants') – substances that give animals and plants the energy essential for growth and life

A decomposer is a living thing that works to break down **organic** materials into smaller parts. For example, a decomposer breaks down fruit on a forest floor by making it rot or decay. The decomposer absorbs the nutrients from the remains and also releases some nutrients into the soil.

Fungi and moulds are decomposers. They decompose plant waste, including vegetables. Some small organisms from the animal kingdom are also decomposers, including worms, slugs, millipedes and termites, which break down logs and sticks. Other decomposers, like maggots and flies, help to break down dead animal carcasses (bodies). Dung (animal droppings) is broken down by dung flies and beetles. Microorganisms, including various bacteria, break down all kinds of substances, including milk, eggs and meat products.

☞ **1** Circle six things that are classified as decomposers.

termite	bacteria	mould	toadstool
maggot	rabbit	worm	eagle

2 Write these words into the spaces below. The lines break them into syllables.

decompose ___ ___ / ___ ___ ___ / ___ ___ ___ ___

decomposers ___ ___ / ___ ___ ___ / ___ ___ ___ / ___ ___ ___

decomposition ___ ___ / ___ ___ ___ / ___ ___ ___ / ___ / ___ ___ ___ ___

3 Complete this passage by filling in the missing vowels (a, e, i, o, u).

D__ c__ mp__ s__ rs break down dead __rg__ n__ sms into sm__ ller parts. Leaf litter, animal d__ ng, bodies of d__ __ d creatures and dead pl__ nts all d__ c__ mp__ s__. After decomposition, the n__ trients that were contained in the decayed organisms return to the s__ __ l. In this way, they are r__ cycl__ d. Nutrients in the soil are then taken up into y__ __ ng plants and __n__ mals eat the plants. The cycle of life and death begins again.

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What is a decomposer? CONT'D

4 Match the decomposer with the material it works on by drawing a line between the two.

Material	Decomposer
a) leaf litter	dung flies
b) logs	slugs
c) dung	lactobacteria
d) fruit	maggots
e) dead animal carcass	termites
f) vegetables	fungi
g) milk	fruit flies

In an ecosystem, the substances contained in food are cycled through the system by producers, consumers and decomposers – although the substances change their form. The main chemical substance in food and living things is called **carbon**.

5 Unscramble the words in brackets to complete the passage.

Dead leaves, grass and (ruitf) _____ and (egetablesv) _____ begin to (ecayd) _____ and become slimy and (mellys) _____ on the ground. (ecomposersD) _____ act on these items in the (armw) _____, (arkd) _____ soil. These decomposers include (acteriab) _____, (ouldm) _____ and (ungif) _____. They (reakb) _____ down the material and the (utrientsn) _____ return to the soil. Nutrients are made of (hemicalsc) _____ that living things need to eat and (bsorba) _____ to stay healthy.

(anureM) _____ and (ottenr) _____ (eatm) _____ from the (arcassesc) _____ of dead animals are another source of food for decomposers. They break these down and the (arbonc) _____ cycle continues in the (cosysteme) _____.

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Name: _____ Date: _____

Food Webs and Food Chains Worksheet

1 Look at this food chain.



lettuce → greenfly → ladybird → thrush → cat

a What does the arrow mean in a food chain? _____

b Name the producer in the food chain _____

c Name the third trophic level in the food chain. _____

d Name the tertiary consumer in the food chain. _____

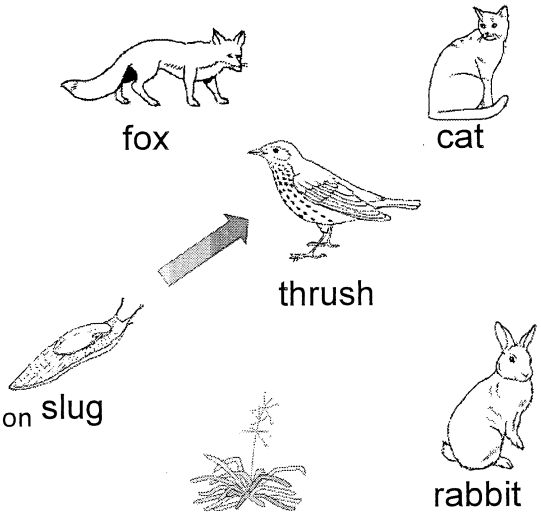
2. Look at these food chains.

grass → slug → thrush → cat

grass → slug → fox

grass → rabbit → fox

grass → rabbit → cat



a Use the food chains to help you fill in the arrows on this food web. One has been added for you.

Use the food web to help you answer these questions.

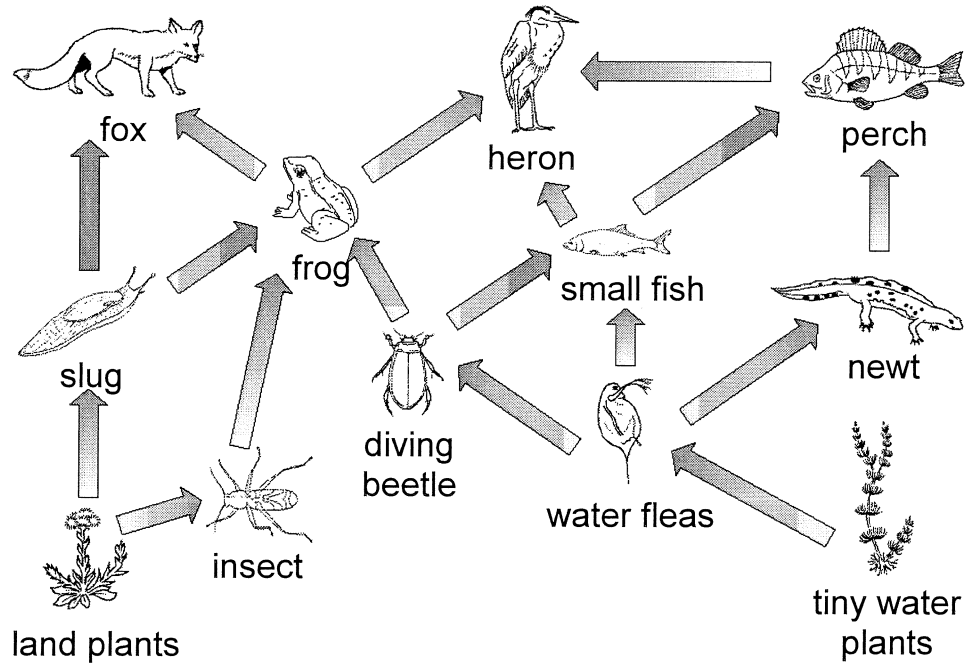
b Name the producer in this food web. _____

c Name a consumer in this food web. _____

d What eats rabbits? _____

e What does a fox eat? _____

3 Look at this food web. Then answer the questions.



a Name two producers in the food web.

_____

b Name three consumers in the food web.

_____

c Write a food chain from this food web with six trophic levels.

_____

d Name the animals that the small fish eats.

_____

e Name the animals that eat the small fish.

_____

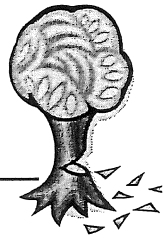
f Explain what could happen to the community if all the frogs suddenly died.

4. How are food webs different to food chains? Explain why food webs are more useful.



Name:_____
Hour:_____

Abiotic vs Biotic Factors



What is the definition of an abotic factor?

_____

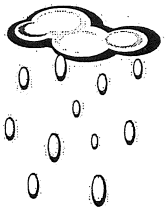
What is the definition of a biotic factor?

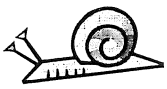
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Enter the items from the following list into a ~~Double Bubble Map~~ T Chart



- Whale
- Mushroom
- Water
- Desert
- Paper
- Glass
- Temperature
- Coral
- Sand
- Clouds
- Snail
- Steak
- Athletes Foot
- Salad
- Mold
- Grass
- Hair
- Ocean
- Tree
- Rocks
- Dirt
- Gold
- Plastic
- Grapes
- Oxygen
- Tundra





BIOTIC | ABIOTIC

Comprehending...

All biotic and abiotic factors are interrelated. In nature you will find that if one factor is changed or removed, it impacts the availability of other resources within the system. Knowing this, give an example of what might happen given the following situations.

In the open space place either an **(A) for abiotic** or **(B) for biotic** to identify the **bolded** object.

1. All of the **rocks**(___) are removed from a desert ecosystem, what would happen to the population of rock dwelling **lizards** (___) and in turn the animals which eat them.

_____

_____

2. A ten mile area of **trees** (___) is removed from the tropical rainforest. How will this affect the amount of **water** (___) and the amount of **oxygen** (___) in the area?

_____

_____

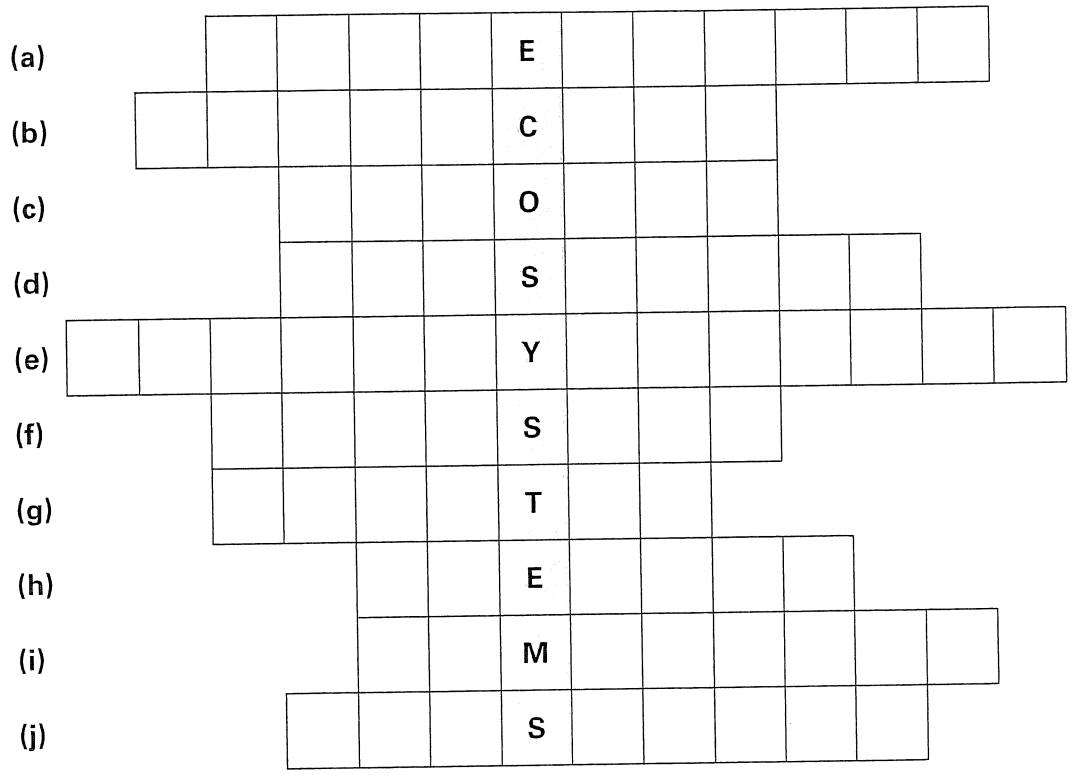
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5.11 Literacy review

Science understanding

FOUNDATION	STANDARD	ADVANCED

1 Write the terms that match the definitions below in the appropriate place in the word grid.



(a) Organisms that have the same food source and live in the same habitat.

(b) Organisms that are able to manufacture their own food.

(c) Non-living factors in the environment are described as this.

(d) Organisms that must eat other organisms to get the energy and nutrients they need.

(e) The process used by plants to make their own food.

(f) An organism that lives on or in a host, taking food or shelter from it.

(g) The place where an organism lives is called this.

(h) The term used to describe a group of living things with common characteristics.

(i) Another name for the interdependence of organisms.

(j) The place where all life as we know it exists.

Describing the environment

11a

1 Read through the information below and circle the correct word in the brackets.

Ecology (is / are) the study of living things and the way they relate to (there / their) environment or surroundings. The environment (include / includes) things like oxygen, (carbon dioxide / carbondioxide), water, (temparature / temperature), soil, plants, animals and even food.

(Organs / Organisms) can occupy different (enviroments / environments). For example, some organisms (live / lived) on land and others live (on / in) water. No matter (were / where) they live, they all interact (for / with) the living and non-living parts of (their / there) environment.

The living parts (off /of) an environment are called (biotic / bionic) and the non-living parts are called abiotic. These non-living and living parts of an environment make up (a / an) ecosystem. Ecosystems can be (vary / very) large like an ocean or very small like a puddle. Each ecosystem, however, (have / has) a range of living things (which / witch) live (in / on) it.

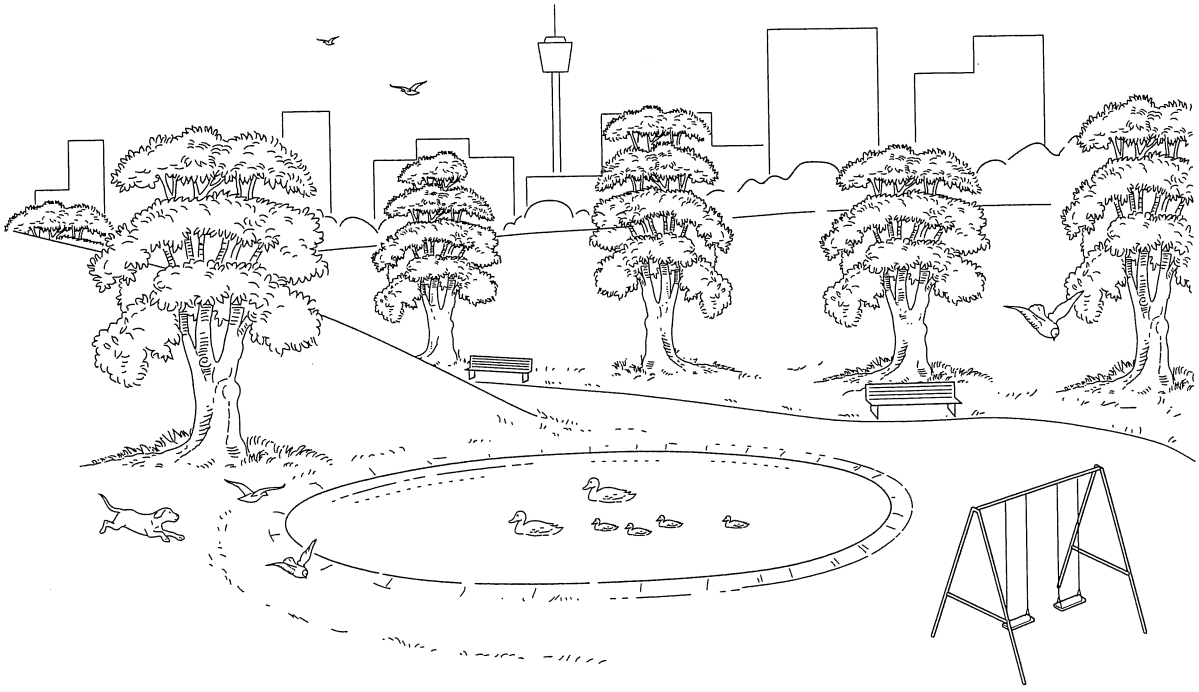
(Within / Without) the ecosystem every living thing has (its/ it's) own special place (to / too) live; this is called its (habitat / habit). The habitat that organisms live (for / in) provides them with all (there / their) needs. A group of organisms (of / off) the same kind, surviving and reproducing amongst each other (in / of) the same habitat, is called a population.

Different groups of organisms that share a habitat make up a (comunity / community). These organisms (need / knead) each other and provide each other (for/ with) shelter and food.

2 Complete the table referring to the text above.

Term	Definition
Abiotic	
Biotic	
Community	
Ecology	
Ecosystem	
Environment	
Habitat	
Population	

3 Observe the diagram below. Classify the things in the diagram as biotic or abiotic. Fill in the table provided to record your observations.



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Describing the environment 11b

Table

4 A group of students in Ms Smart's class visited the park shown in the diagram. They took some measurements in three different areas in the park. Their results are shown in the table below.

Aspect	Area 1	Area 2	Area 3
Temperature (°C)	18	26	22
Humidity (%)	–	30	50
Light intensity (1 (min) → 10 (max))	3	10	6
Wind speed (km/h)	0	11	7

a State which area is:

i) in the pond _____

ii) under the shade of trees _____

iii) in the middle of the oval _____

b Suggest several types of organisms you would expect to see in these three areas.

i) Area 1 ii) Area 2 iii) Area 3

_____ _____ _____

_____ _____ _____

_____ _____ _____

_____ _____ _____

_____ _____ _____

5 Describe the specific biotic and abiotic features from one of the three areas.

_____

_____

_____

6 List what other abiotic things could be measured in the pond ecosystem and suggest possible measurements.

_____

_____

_____

_____

_____

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